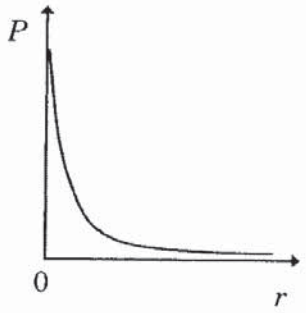
PART II : CHEMISTRY

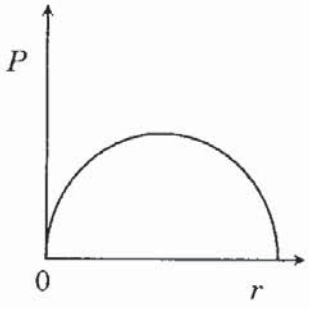
SECTION 1 (Maximum Marks: 15)

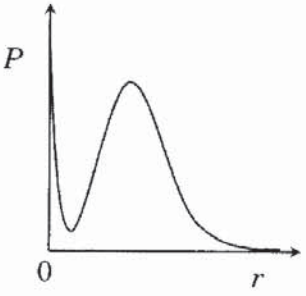
- This section contains **FIVE** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories:

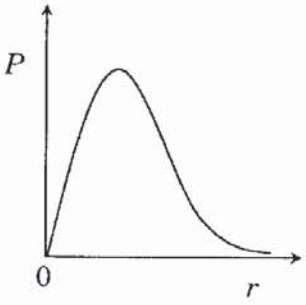
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : –1 In all other cases.

Q.19 P is the probability of finding the 1s electron of hydrogen atom in a spherical shell of infinitesimal thickness, dr, at a distance r from the nucleus. The volume of this shell is $4\pi r^2 dr$. The qualitative sketch of the dependence of P on r is

(A)


(B)


(C)


(D)


Space for rough work

Answers for the above questions

Ans for Q.19: (D)

Q.20 One mole of an ideal gas at 300 K in thermal contact with surroundings expands isothermally from 1.0 L to 2.0 L against a constant pressure of 3.0 atm. In this process, the change in entropy of surroundings (ΔS_{surr}) in J K^{-1} is

(1 L atm = 101.3 J)

(A) 5.763 (B) 1.013 (C) −1.013 (D) −5.763

Q.21 The increasing order of atomic radii of the following Group 13 elements is

(A) Al < Ga < In < Tl (B) Ga < Al < In < Tl

(C) Al < In < Ga < Tl (D) Al < Ga < Tl < In

Q.22 Among $[Ni(CO)_4]$, $[NiCl_4]^{2-}$, $[Co(NH_3)_4Cl_2]Cl$, $Na_3[CoF_6]$, Na_2O_2 and CsO_2, the total number of paramagnetic compounds is

(A) 2 (B) 3 (C) 4 (D) 5

Q.23 On complete hydrogenation, natural rubber produces

(A) ethylene-propylene copolymer (B) vulcanised rubber

(C) polypropylene (D) polybutylene

Space for rough work

Answers for the above questions

Ans for Q.20: (C)	Ans for Q.21: (B)	Ans for Q.22: (B)	Ans for Q.23: (A)

SECTION 2 (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS.
- For each question, marks will be awarded in one of the following categories:

Full Marks	: +4	If only the bubble(s) corresponding to all the correct option(s) is(are) darkened.
Partial Marks	: +1	For darkening a bubble corresponding **to each correct option**, provided NO incorrect option is darkened.
Zero Marks	: 0	If none of the bubbles is darkened.
Negative Marks	: –2	In all other cases.

- For example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in +4 marks; darkening only (A) and (D) will result in +2 marks; and darkening (A) and (B) will result in –2 marks, as a wrong option is also darkened.

Q.24 According to the Arrhenius equation,

(A) a high activation energy usually implies a fast reaction.

(B) rate constant increases with increase in temperature. This is due to a greater number of collisions whose energy exceeds the activation energy.

(C) higher the magnitude of activation energy, stronger is the temperature dependence of the rate constant.

(D) the pre-exponential factor is a measure of the rate at which collisions occur, irrespective of their energy.

Q.25 A plot of the number of neutrons (N) against the number of protons (P) of stable nuclei exhibits upward deviation from linearity for atomic number, $Z > 20$. For an unstable nucleus having N/P ratio less than 1, the possible mode(s) of decay is(are)

(A) β^--decay (β emission)

(B) orbital or K-electron capture

(C) neutron emission

(D) β^+-decay (positron emission)

Space for rough work

Answers for the above questions

Ans for Q.24: (B), (C) and (D)	Ans for Q.25: (B) and (D)

Q.26 The crystalline form of borax has

(A) tetranuclear $[B_4O_5(OH)_4]^{2-}$ unit

(B) all boron atoms in the same plane

(C) equal number of sp^2 and sp^3 hybridized boron atoms

(D) one terminal hydroxide per boron atom

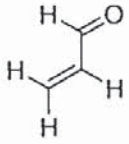
Q.27 The compound(s) with TWO lone pairs of electrons on the central atom is(are)

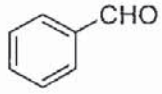
(A) BrF_5 (B) ClF_3 (C) XeF_4 (D) SF_4

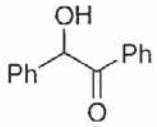
Q.28 The reagent(s) that can selectively precipitate S^{2-} from a mixture of S^{2-} and SO_4^{2-} in aqueous solution is(are)

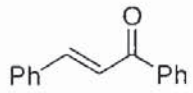
(A) $CuCl_2$ (B) $BaCl_2$ (C) $Pb(OOCCH_3)_2$ (D) $Na_2[Fe(CN)_5NO]$

Q.29 Positive Tollen's test is observed for

(A) 

(B) 

(C) 

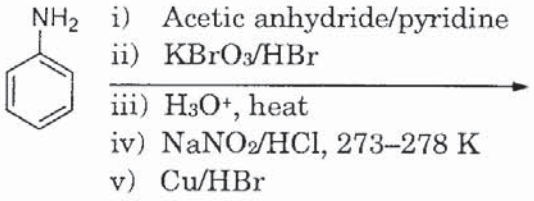
(D) 

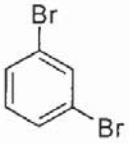
Space for rough work

Answers for the above questions

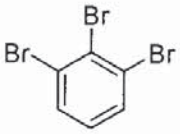
Ans for Q.26: (A), (C) and (D)	**Ans for Q.27: (B) and (C)**
Ans for Q.28: (A) OR (A) and (C)	**Ans for Q.29: (A), (B) and (C)**

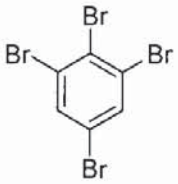
Q.30 The product(s) of the following reaction sequence is(are)



(A) 

(B) 

(C) 

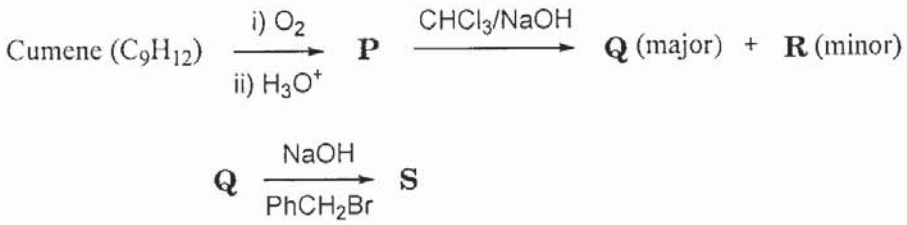
(D) 

Space for rough work

Answers for the above questions

Ans for Q.30: (B)

Q.31 The correct statement(s) about the following reaction sequence is(are)



(A) **R** is steam volatile

(B) **Q** gives dark violet coloration with 1% aqueous $FeCl_3$ solution

(C) **S** gives yellow precipitate with 2, 4-dinitrophenylhydrazine

(D) **S** gives dark violet coloration with 1% aqueous $FeCl_3$ solution

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Space for rough work

Answers for the above questions

Ans for Q.31: (B) and (C)

SECTION 3 (Maximum Marks: 15)

- This section contains **FIVE** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories:

Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.

Zero Marks : 0 In all other cases.

Q.32 The mole fraction of a solute in a solution is 0.1. At 298 K, molarity of this solution is the same as its molality. Density of this solution at 298 K is 2.0 g cm^{-3}. The ratio of the molecular weights of the solute and solvent, $\left(\frac{MW_{solute}}{MW_{solvent}}\right)$, is

Q.33 The diffusion coefficient of an ideal gas is proportional to its mean free path and mean speed. The absolute temperature of an ideal gas is increased 4 times and its pressure is increased 2 times. As a result, the diffusion coefficient of this gas increases x times. The value of x is

Space for rough work

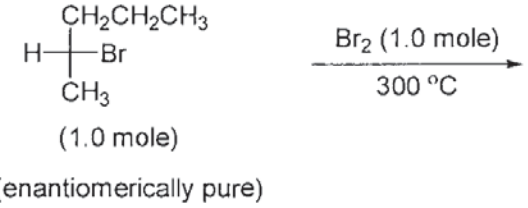
Answers for the above questions

Ans for Q.32: (9)	Ans for Q.33: (4)

Q.34 In neutral or faintly alkaline solution, 8 moles of permanganate anion quantitatively oxidize thiosulphate anions to produce **X** moles of a sulphur containing product. The magnitude of **X** is

Q.35 The number of geometric isomers possible for the complex $[CoL_2Cl_2]^-$ $(L = H_2NCH_2CH_2O^-)$ is

Q.36 In the following monobromination reaction, the number of possible chiral products is



(1.0 mole)

(enantiomerically pure)

END OF PART II : CHEMISTRY

Space for rough work

Answers for the above questions

Ans for Q.34: (6)	Ans for Q.35: (5)	Ans for Q.36: (5)